Reward Recovery in Sequential Decision Models

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1 Setup and Notation

Consider a stationary infinite-horizon MDP with finite state space $\mathcal{S}$, finite action space $\mathcal{A}$, discount factor $\beta \in (0,1)$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, and unknown reward $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. Additive T1EV shocks yield the soft Bellman recursion.

$$Q^*(s,a) = r(s,a) + \beta \sum_{s'} P(s' \mid s,a)\, V^*(s'), \tag{1}$$

$$V^*(s) = \log \sum_{a'} \exp Q^*(s,a'), \tag{2}$$

$$\pi^*(a \mid s) = \exp\big(Q^*(s,a) - V^*(s)\big), \tag{3}$$

$$A^*(s,a) = Q^*(s,a) - V^*(s) = \log \pi^*(a \mid s). \tag{4}$$

The advantage A^* is identified from CCPs. The reward r is not: any $\tilde{r}(s,a) = r(s,a) + \lambda(s)$ generates identical CCPs (Magnac and Thesmar, 2002). Following Kalouptsidi et al. (2021), call the set of all such $\tilde{r}$ the identified set.

2 Four Approaches to Reward Recovery

2.1 Reduced-Form Q

The reduced-form approach estimates $Q_\omega(s,a)$ by logit MLE.

$$\hat{\omega} = \arg\max_{\omega} \sum_{i,t} \log \frac{\exp Q_\omega(s_{it}, a_{it})}{\sum_{a'} \exp Q_\omega(s_{it}, a')}. \tag{5}$$

At the population optimum, $Q_{\hat{\omega}}(s,a) = Q^*(s,a) + c(s)$ for some action-independent c, so the backed-out "reward" is the advantage $A^*(s,a)$, not the true reward $r(s,a)$.

2.2 AIRL (Fu et al., 2018)

The AIRL discriminator decomposes as follows.

$$f_\theta(s,a,s') = g_\theta(s,a) + \beta\, h_\phi(s') - h_\phi(s), \tag{6}$$

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Here g_θ approximates the reward and $\beta h(s') - h(s)$ is potential-based shaping (Ng et al., 1999). At the GAN optimum the discriminator score equals the advantage, $f^* = A^*$ (Fu et al., 2018, Appendix A.4), which implies the following expression for the recovered reward network.

$$g^*(s,a) = r(s,a) - \delta(s) + \beta \sum_{s'} P(s' \mid s,a)\, \delta(s'), \tag{7}$$

where $\delta(s) \equiv V^*(s) - h^*(s)$. Unless $\delta \equiv 0$, the recovered g^* is a potential-based perturbation of r.

2.2.1 LSW Anchor Identification (Lee et al., 2026)

LSW impose two anchors. First, the exit payoff is normalized to zero, so $g(s, \text{exit}) = 0$ for all s. Second, the shaping function is pinned at the absorbing state, so $h(s_{\text{abs}}) = 0$, where exit deterministically transitions to s_{abs}.

Lemma 1 (Non-identifiability)**.** *If (g,h) and $(\bar{g},\bar{h})$ produce the same f for all (s,a,s'), then $\bar{h} = h + \Phi$ and $\bar{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s')$ for some $\Phi : \mathcal{S} \to \mathbb{R}$.*

Proof. Rearrange $g + \beta h(s') - h(s) = \bar{g} + \beta\bar{h}(s') - \bar{h}(s)$ and define $\Phi = \bar{h} - h$. □ □

Theorem 2 (Anchor identification)**.** *Under (A1)–(A2), if (g,h) and $(\bar{g},\bar{h})$ realize the same f, then $g = \bar{g}$ and $h = \bar{h}$.*

Proof. By Lemma 1, $\bar{g}(s, \text{exit}) = 0 + \Phi(s) - \beta\Phi(s_{\text{abs}})$. Anchor (A1) forces $\Phi(s) = \beta\Phi(s_{\text{abs}})$ for all s. Anchor (A2) forces $\Phi(s_{\text{abs}}) = 0$. Hence $\Phi \equiv 0$. □ □

Theorem 3 (Recovery of true reward)**.** *If the true pair (r^*, V^*) satisfies $r^*(s, \text{exit}) = 0$ and $V^*(s_{\text{abs}}) = 0$, then any AIRL pair (g,h) realizing A^* under the same anchors satisfies $g = r^*$ and $h = V^*$.*

Proof. Define $\Phi = h - V^*$. The exit anchor gives $\Phi(s) = \beta\Phi(s_{\text{abs}})$. The absorbing anchor applied to both h and V^* gives $\Phi(s_{\text{abs}}) = 0$. □ □

The Kang et al. (2025) ERM framework uses the same anchor principle (fixing the reward of a known action) but solves a single non-adversarial optimization.

2.3 IQ-Learn (Garg et al., 2021)

The inverse Bellman operator $r_Q(s,a) \equiv Q(s,a) - \beta \sum_{s'} P(s' \mid s,a)\, V_Q(s')$ expresses reward as a function of Q alone. IQ-Learn estimates Q by solving the following optimization problem.

$$\max_Q \; \mathbb{E}_{(s,a)\sim\rho^*}\big[\phi\big(r_Q(s,a)\big)\big] - (1-\beta)\, \mathbb{E}_{s_0\sim\mu_0}\big[V_Q(s_0)\big], \tag{8}$$

The function $\phi(x) = x - x^2/2$ corresponds to χ^2 divergence. At the optimum, r_Q separates immediate reward from continuation values via the Bellman constraint.

2.4 GLADIUS (Kang et al., 2025)

GLADIUS parameterizes both $Q_\theta(s,a)$ and $\hat{E}_\theta[V(s') \mid s,a]$ with neural networks, trained jointly on observed (s,a,s') transitions:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_i \log \pi_\theta(a_i \mid s_i) + \lambda \frac{1}{N}\sum_i \big(\hat{E}_\theta[V(s_i') \mid s_i, a_i] - V_\theta(s_i')\big)^2, \tag{9}$$

where $V_\theta(s') = \sigma \log \sum_{a'} \exp(Q_\theta(s',a')/\sigma)$. After training, reward is extracted via the same inverse Bellman operator as IQ-Learn: $\hat{r}(s,a) = Q_\theta(s,a) - \beta\, \hat{E}_\theta[V(s') \mid s,a]$. The MLP architecture shares parameters across states, providing implicit regularization over tabular Q.

3 Counterfactual Taxonomy

Definition 4 (Type I: state extrapolation)**.** *The transition kernel P and reward r are unchanged. Only the realized state values shift. The counterfactual CCPs require only evaluating the estimated Q at new state inputs, with no Bellman re-solve.*

Definition 5 (Type II: environment change)**.** *The reward r is unchanged, but the transition kernel shifts to $\tilde{P}$ or the action set changes. The counterfactual Q must satisfy a new Bellman equation $\tilde{Q} = r + \beta\tilde{P}\tilde{V}$, which requires the structural reward r separated from the old continuation values.*

Proposition 6 (Type I equivalence)**.** *Under Type I, all four methods yield identical counterfactual CCPs.*

Proof. Each recovers $Q^*(s,a) + c(s)$ for some action-independent c. The softmax $\pi(a \mid \tilde{s}) \propto \exp(Q^*(\tilde{s},a) + c(\tilde{s}))$ cancels c. □ □

Proposition 7 (Type II divergence)**.** *Under Type II, reduced-form produces invalid predictions. AIRL with anchors, IQ-Learn, and GLADIUS produce valid predictions when r is correctly recovered.*

The reduced-form estimate $Q^* + c(s)$ bundles r with $\beta P V^*$. Under $\tilde{P}$, continuation values change but the reduced-form has no way to disentangle r. AIRL with anchors recovers $g^* = r^*$ exactly (Theorem 3). IQ-Learn and GLADIUS separate r via the inverse Bellman operator. The LSW anchors provide what Kalouptsidi et al. (2021) call the "additional restrictions" needed to identify counterfactuals under transition changes.

4 Application to Lee, Sudhir & Wang (2026)

LSW study sequential content consumption with three actions (buy, wait, exit) using AIRL under anchors (A1)–(A2) and deterministic transitions. Their wait-time validation (Section 6.3) is a Type I counterfactual, so a reduced-form logit would pass this test identically. Their WFF removal and segment-specific delay policies (Sections 7.1–7.2) are Type II because they change the action-set structure and the MDP that each segment faces. Their content-based paywall (Section 7.3) is also Type II because high-content episodes become paid-only, altering the transition structure episode-by-episode.

5 Simulation Study

We construct a tabular MDP that mimics the LSW book-reading environment. The state encodes episode position, content quality, and wait-time level, giving $15 \times 3 \times 4 = 180$ regular states plus one absorbing state. The three actions are buy, wait, and exit, with deterministic transitions and $\beta = 0.95$. The true reward is $r^*(s, \text{buy}) = u(c) - 2.0$, $r^*(s, \text{wait}) = u(c) - 0.08w$, and $r^*(s, \text{exit}) = 0$, where $u \in \{0.5, 1.5, 3.0\}$. Analyses A through C and E through F are evaluated at the population level with no estimation noise. Analysis D uses finite-sample estimation.

5.1 A. Reward recovery

Methods with anchors pin the value function offset and recover r^* exactly. Reduced-form logit converges to the advantage $A^* = r^* + \beta PV^* - V^*$, which is a noisy proxy for the reward because V^* adds large state-specific offsets.

Method	MSE vs. r^*	Corr. with r^*
Oracle / AIRL+anchors / IQ / GLADIUS	0.000	1.000
AIRL-no-anchors	10.005	0.440
Reduced-form	40.020	0.138

Table 1: Population-level reward recovery. Type I CCP errors are below 10^{-6} for all methods.

5.2 B. Type II error vs. transition shift

The counterfactual changes the transition kernel so that buying advances k episodes instead of one. The Bellman equation is re-solved under each method's recovered reward.

k	Oracle/AIRL+a/IQ/GLADIUS	AIRL-no-a	Reduced-form
1	0.000	0.000	0.000
2	0.000	0.025	0.082
3	0.000	0.017	0.094
5	0.000	0.015	0.123
7	0.000	0.016	0.129
10	0.000	0.014	0.139

Table 2: Type II CCP error. Structural methods are exact. Reduced-form error grows with shift magnitude.

5.3 C. Shaping magnitude sweep

We construct shaped rewards $\tilde{r} = r^* + \alpha V^*(s) - \alpha\beta V^*(s')$ for $\alpha \in [0, 1]$ and evaluate the Type II counterfactual with $k = 3$.

α	Type II CF error	Corr. with r^*
0.00	0.000	1.000
0.10	0.004	0.908
0.30	0.011	0.610
0.50	0.017	0.440
1.00	0.027	0.270

Table 3: CF error is monotone in shaping magnitude. Without anchors, α is indeterminate.

5.4 D. Finite-sample convergence

The tabular χ^2 IQ-Learn objective has multiple local optima in finite samples. We compare three finite-sample estimators. The reduced-form logit converges to A^*, which is the wrong target for Type II counterfactuals. The anchor-constrained MLE adds soft penalties on the implied reward at exit and absorbing states to the logit NLL. GLADIUS uses neural Q and EV networks.

N	Reduced-form	Anchor-MLE	GLADIUS
500	0.264	0.255	0.106
2,000	0.262	0.255	0.123
10,000	0.262	0.257	0.099

Table 4: Type II CF error ($k = 3$, 2 seeds). GLADIUS cuts error by 60% over tabular methods via implicit neural regularization.

5.5 E. Anchor misspecification

We set the true exit payoff to ε while the anchors assume it is zero.

ε	Corr. with r^*	Type II CF error	In-sample π shift
0.0	1.000	0.000	0.000
0.5	0.976	0.002	0.000
1.0	0.909	0.004	0.000
2.0	0.729	0.008	0.000

Table 5: Degradation is smooth, not knife-edge. In-sample CCPs are unaffected.

5.6 F. Content-based paywall

High-content episodes become paid-only and the wait action is blocked, forcing buy-or-exit. This is a Type II shift that mirrors Section 7.3 of LSW.

Method	CCP error
Oracle / AIRL+anchors / IQ-Learn	0.000
AIRL-no-anchors	0.009
Reduced-form	0.185

Table 6: Reduced-form fails at 18.5pp because it cannot separate content utility from continuation value.

6 Summary

	Reduced-form	AIRL (generic)	AIRL (anchors)	IQ-Learn / GLADIUS
Recovers	$A^*(s,a)$	r + shaping	$r^*(s,a)$	$r^*(s,a)$
Type I CF	✓	✓	✓	✓
Type II CF	×	Only if $\delta \equiv 0$	✓	✓

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